DAY 4 · LECTURE & DEMO · 90 MINUTES

The Admiral's Arsenal

Cloud-Native Classrooms: vCluster, KubeVirt & Chaos

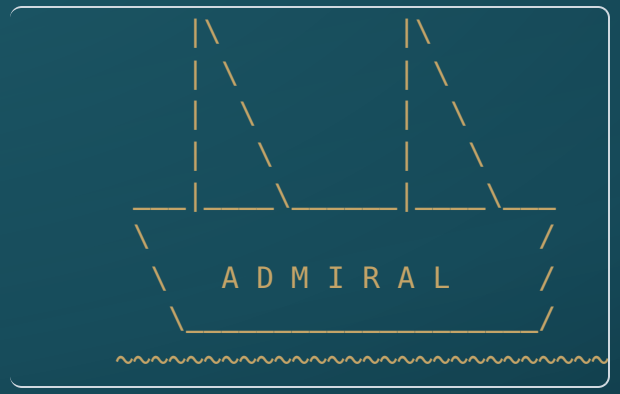
Three days of building. This morning, the most advanced manoeuvres in the fleet.

Where we are

- **Days 1–3**: you built, connected, templated, and automated — container images, live Pods, three-tier deployments, GitOps pipelines.
- This morning, Admiral Bash opens the arsenal: the tools that make *this whole curriculum portable to your own college*.
- Four parts — three tools, one bridge to the lab:
  - **Part I** — The Bunk and the Ship: `vCluster`
  - **Part II** — The Legacy Fleet: `KubeVirt`
  - **Part III** — Summoning the Storm: Chaos Engineering
  - **Part IV** — The Ultimate Exam (+ what comes next)
- Then the break — and the Pirate strikes.

PART I

The Bunk and the Ship



> *"What a ship is... is freedom." — Captain Jack Sparrow, Pirates of the Caribbean: The Curse of the Black Pearl*

The problem with namespaces

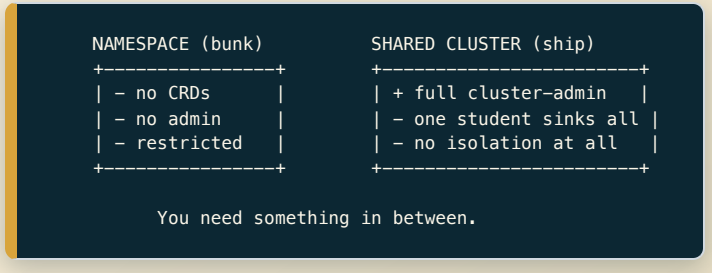
- So far, each student gets their own **namespace** — their own patch of the shared ocean.
- A namespace is like a **bunk on a ship**: private, but the ship's rules still govern everything.
- Students **cannot**:
  - install Custom Resource Definitions (CRDs)
  - create cluster-level resources
  - touch anything outside their namespace
- For most labs, that is fine. For advanced courses, it is a cage.

The other extreme: cluster-admin

- Give a student `cluster-admin` on the shared cluster and they have the keys to the whole ship.
- One misfired `kubectl delete` wipes resources belonging to 29 other students.
- One runaway process drains the cluster's CPU for the whole room.
- One mistake during a CRD exercise breaks the control plane for everyone.

> *Sharing `cluster-admin` on a live teaching cluster is not a risk — it is a when, not an if.*

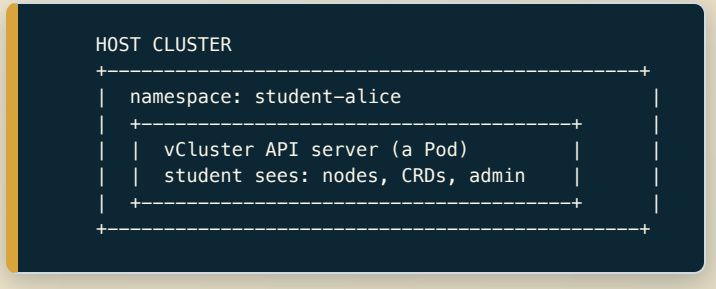
The bunk is too small. The ship is too dangerous.



- Students need the *feeling* of cluster-admin without the blast radius.
- The answer: give each student their **own Kubernetes cluster** — without buying 30 clusters.

vCluster: a cluster inside a namespace

- **vCluster** deploys a full, isolated Kubernetes API server *inside* a single namespace on the host cluster.
- From the student's perspective: a real multi-node cluster, full `cluster-admin`, CRDs welcome.
- From the host cluster's perspective: one namespace, one set of Pods — ordinary traffic.
- The host cluster never sees the student's internal objects; the student never touches the host.



vCluster: what the instructor controls

- **Spin up**: one Helm install per student — or scripted for 30 at once.
- **Tear down**: `helm uninstall` — the whole cluster disappears with it.
- **Isolation**: a student cannot escape their vCluster into the host.
- **Resource limits**: the namespace wrapping the vCluster can have CPU/memory quotas.
- Students get the *experience* of owning a cluster; the instructor keeps the keys to the ship.

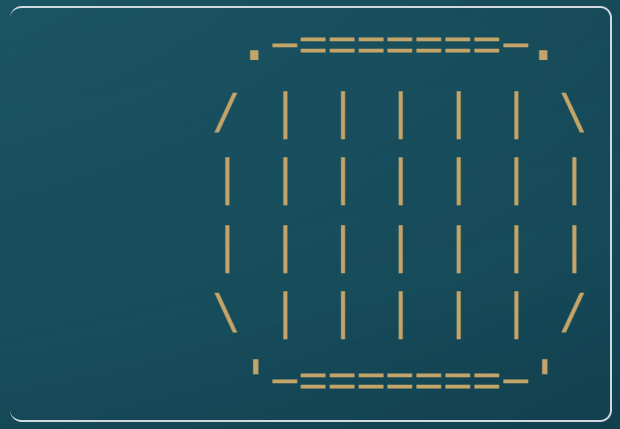
vCluster: the classroom payoff

- **30 students, 30 real clusters** — on one shared host, no provisioning from IT.
- Students can install operators, experiment with CRDs, break things freely.
- Advanced courses — service mesh, multi-tenant SaaS, operator development — all become teachable without a cloud bill per student.
- When a student melts their cluster: `helm uninstall` + `helm install` — reset in under a minute.

This is what it looks like when IT is no longer the bottleneck on what you can teach.

PART II

The Legacy Fleet



> *"Don't scuttle the old fleet just because ye launched a new one. Some cargoes need the old hull." — the Boatswain*

The hesitation every faculty member has

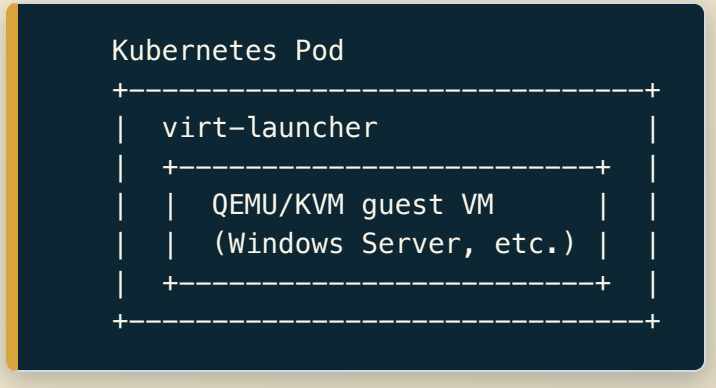
- "This is great — but I still have to teach **Windows Server**."
- "My program covers **Active Directory**, DHCP, legacy Linux admin."
- "I can't abandon that curriculum just because Kubernetes is shiny."

This is a legitimate structural constraint, not a failure of imagination.

A Kubernetes-only lab environment looks like a step backward if you still owe your students Windows Server administration.

KubeVirt: VMs as Kubernetes objects

- **KubeVirt** is a CNCF project that adds a new resource type to Kubernetes: the `VirtualMachine`.
- Under the hood, it wraps a standard **QEMU/KVM** virtual machine inside a Kubernetes Pod.
- The VM boots like any desktop VM — Windows, Linux, whatever image you supply.
- From Kubernetes' point of view: just another Pod to schedule, watch, and restart.



KubeVirt: demo — a VM on the cluster

Live demo — instructor projects terminal.

- Apply a `VirtualMachine` manifest to the cluster.
- Start it: `virtctl start my-windows-vm`
- Console in: `virtctl console my-windows-vm`

The VM is live on the **same virtual network** as the Day 2 web Pods.

```
ping from windows-vm --> frontend-pod    (it works)
curl from backend-pod --> windows-vm     (it works)
```

One cluster. Containers and VMs sharing the same network fabric.

KubeVirt: the classroom payoff

- Teach Windows Server administration and containerized microservices in the **same lab session**.
- Students configure Active Directory — then connect it to a containerized app — in one unified environment.
- No separate VM infrastructure. No separate network. No separate IT queue.
- Hardcore IT programs use this to move fully onto Kubernetes **without throwing away** a decade of legacy lab content.

Legacy curriculum	Kubernetes curriculum
Windows Server	containerized apps
Active Directory	Kubernetes RBAC
DHCP / DNS	CoreDNS + Services
All on the **same cluster**	← same cluster

PART III

Summoning the Storm

> *"A man can be destroyed but not defeated." — Ernest Hemingway, The Old Man and the Sea*

Real systems fail

- No production system runs forever without a fault.
- The question is not *whether* it will fail — it is *whether you find out* from your own monitoring, or from a flood of angry users.
- **Chaos Engineering** means intentionally breaking your own system — under controlled conditions, in your own lab — to *prove* it can survive.
- This is not recklessness. It is the methodology of modern **Site Reliability Engineering (SRE)**.

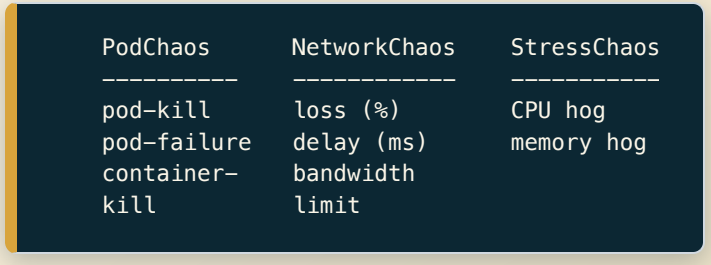
> "*Hope is not a reliability strategy.*" — *SRE axiom*

The tools: Chaos Mesh and LitmusChaos

- **Chaos Mesh** — CNCF-hosted. Chaos experiments are Kubernetes CRDs: apply a manifest, the chaos begins; delete it, the storm stops.
- **LitmusChaos** — CNCF-hosted. Workflow-oriented chaos with built-in dashboards and test reports.
- Both run *inside* the cluster — no external attack infrastructure needed.
- Both define faults as ordinary Kubernetes objects — inspectable, version-controlled, shareable.

*For this seminar we use **Chaos Mesh**. The experiments you will see this morning are already staged.*

The kinds of faults



- **PodChaos** — violently kill a Pod (or container) on a schedule.
- **NetworkChaos** — drop a percentage of packets, inject latency, cap bandwidth — *between specific namespaces*.
- **StressChaos** — flood CPU or memory inside a running container.

All are scoped with a **selector**: target by namespace, label, or Pod name.

What a chaos manifest looks like

```yaml
apiVersion: chaos-mesh.org/v1alpha1
kind: NetworkChaos
metadata:
  name: kraken-packet-loss
  namespace: chaos
spec:
  action: loss
  mode: all
  duration: "90m"
  loss:
    loss: "40"
  selector:
    namespaces: ["blackbeard", "annebonny"]
```

- One manifest. Drops **40%** of packets in both directions for 90 minutes.
- Apply it: the storm begins. Delete it: instant calm.
- Students can *see* this object — that visibility is core to the exercise.

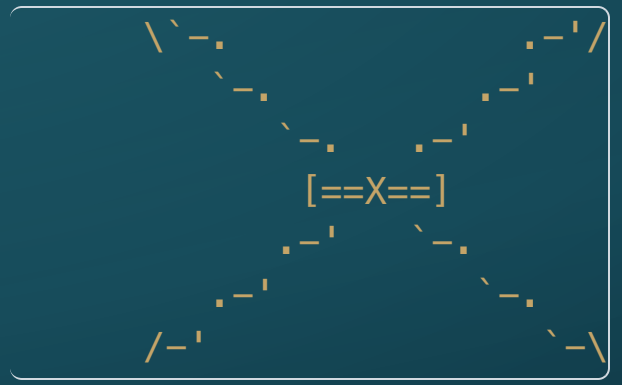
Chaos Engineering as SRE practice

- **Netflix Simian Army** coined the practice — randomly terminate production instances to prove resilience.
- **Google SRE** runs "DiRT" (Disaster Recovery Training) — planned, scoped chaos against live services.
- The insight: a system that is never tested under failure is a system whose failure mode is unknown.
- **Your curriculum already has the scaffolding**: Deployments auto-respawn Pods; ArgoCD auto-heals GitOps drift. The students built a resilient system *without knowing it was resilient*.

After the break, they will find out.

PART IV

The Ultimate Exam



> *"A smooth sea never made a skilled sailor." — a sailor's proverb*

Chaos as assessment

- A traditional exam asks: *"Do you know what a NetworkPolicy is?"*
- A chaos exam asks: *"Fix the live system. It is losing 40% of its packets. Go."*
- The `NetworkChaos` manifest **is** the test paper.
- The student's `kubectl get events` output **is** the answer sheet.
- You grade the **recovery**, not the recall.

This is the most authentic assessment a DevOps / systems administration program can run.

What makes it gradeable

What you observe	What it proves
`kubectl get events` before panicking	systematic diagnosis
Finding the `NetworkChaos` object	connecting symptoms to cause
Writing a `NetworkPolicy` via GitOps	disciplined response under pressure
60 s of clean traffic while attack runs	actual remediation

- No rubric guessing. No "partial credit for knowing the command."
- The system either serves traffic reliably or it does not.

The bridge: what this morning built toward

This morning's three topics are not independent demos. They form a stack:

```
vCluster    -> safe, full clusters for every student
KubeVirt    -> legacy curriculum lives on the same platform
Chaos Mesh  -> authentic, self-grading assessment built in
```

- Together they are the answer to: "*How do I run this curriculum at my college?*"
- After the break, you will not just understand Chaos Mesh — you will be inside a live attack.

> **Foreshadow:** *during the 10:30 break, the Pirate boards the cluster. When you return, something will already be wrong. That is not a drill.*

Instructor Superpower

THE CLOUD-NATIVE CLASSROOM STACK

This morning's three tools all point the same way — at what you can now build yourself:

- **vCluster** — 30 students, 30 real clusters on one shared host. Stand it up yourself.
- **KubeVirt** — teach Windows Server and legacy Linux admin on the same unified cluster as your container labs.
- **Chaos Mesh** — turn a `NetworkChaos` manifest into an auto-gradeable capstone exam.
- The environments you can choose to support have radically expanded. You don't wait for IT to say "sure, we can support that" — you build it, for a lesson, a course, or a seminar like this one.

Up next: Lab 01 — The Pirate Strikes

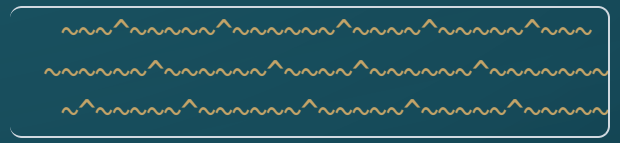
- The cluster is about to come **under attack**.
- Pods in your namespace will die and respawn. Your 3-tier pipeline will flicker.
- A Pirate has boarded — running live **Chaos Mesh** experiments against the fleet.
- Your mission: **trace the root cause and stabilize your logistics pipeline** while the attack is still running.

What you will use:

- `k9s` — watch the carnage in real time
- `kubectl get events` — read the ship's incident log
- `kubectl get podchaos,networkchaos -A` — name your enemy
- `NetworkPolicy` via **GitOps** — raise the blockade

> *After the break, the storm is already running. Do not panic — observe first.*

To the storm.



The arsenal is loaded. The Pirate is coming. Prove the fleet is ready.